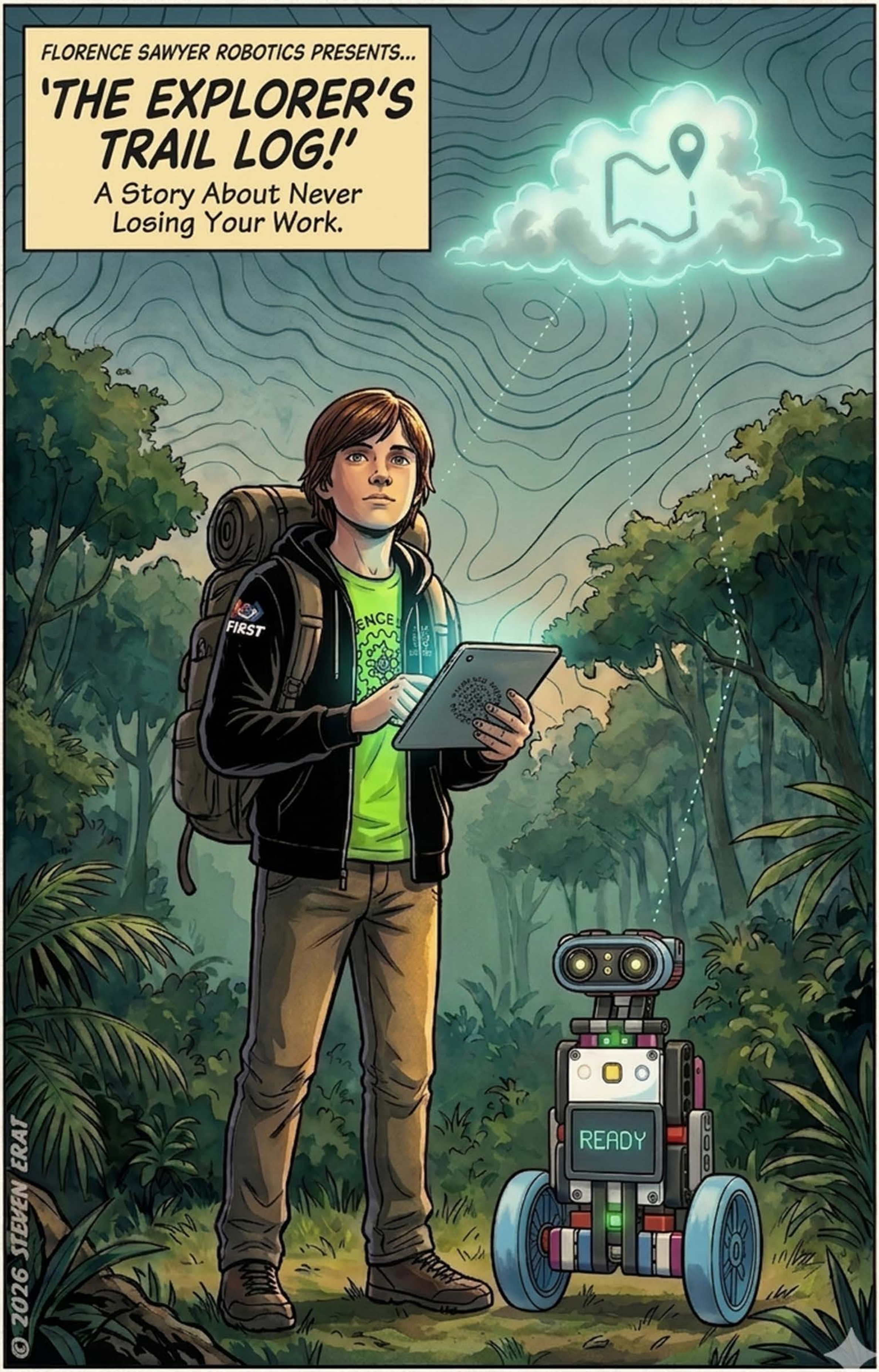
FLORENCE SAWYER ROBOTICS PRESENTS...
'THE EXPLORER'S TRAIL LOG!'
A Story About Never Losing Your Work.
FIRST
READY
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CHAPTER ONE: NEVER TRUST A SINGLE PIECE OF GEAR

The code is GONE. Jack has the thumb drive — at home!

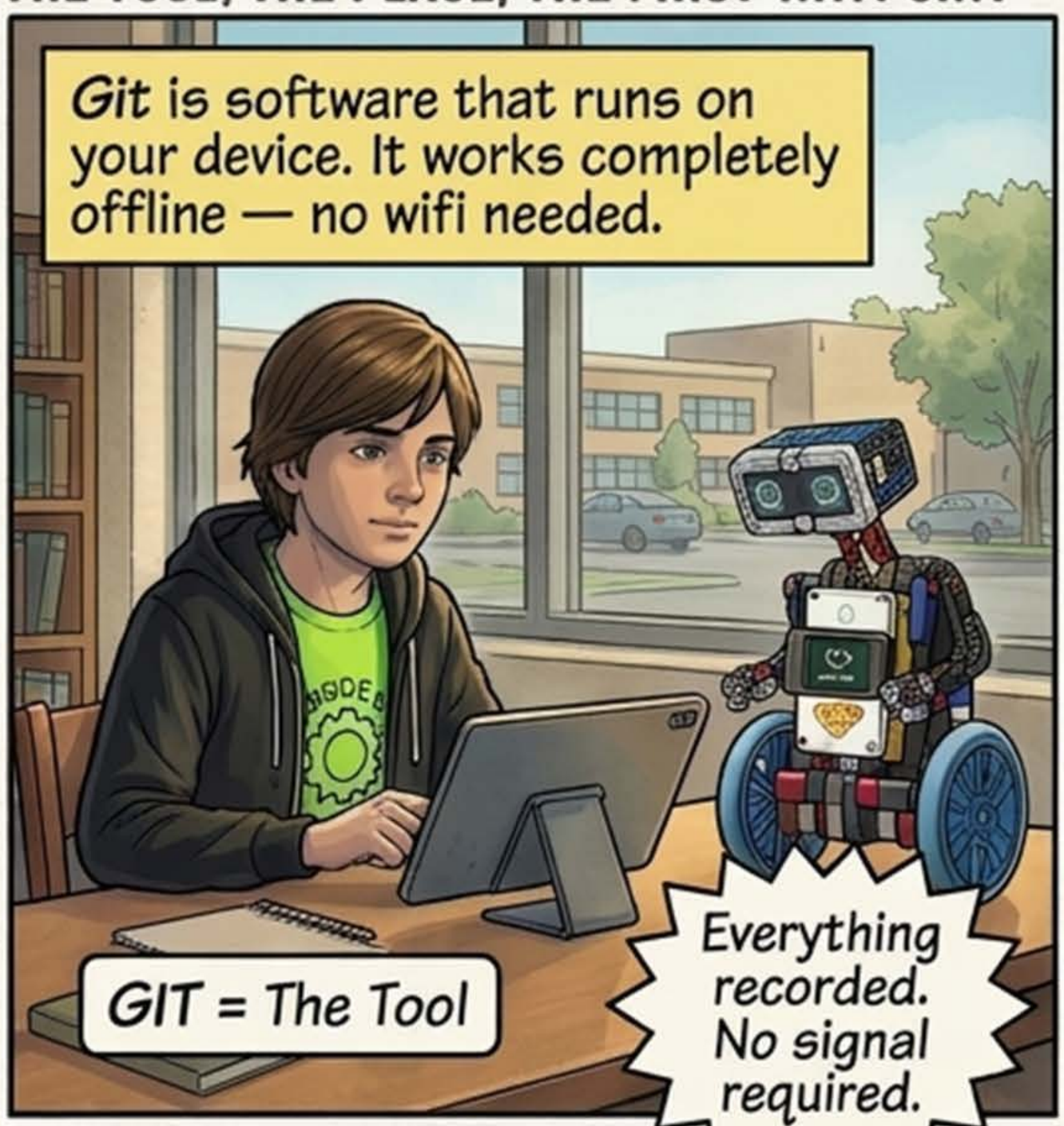
Competition is in THREE DAYS!

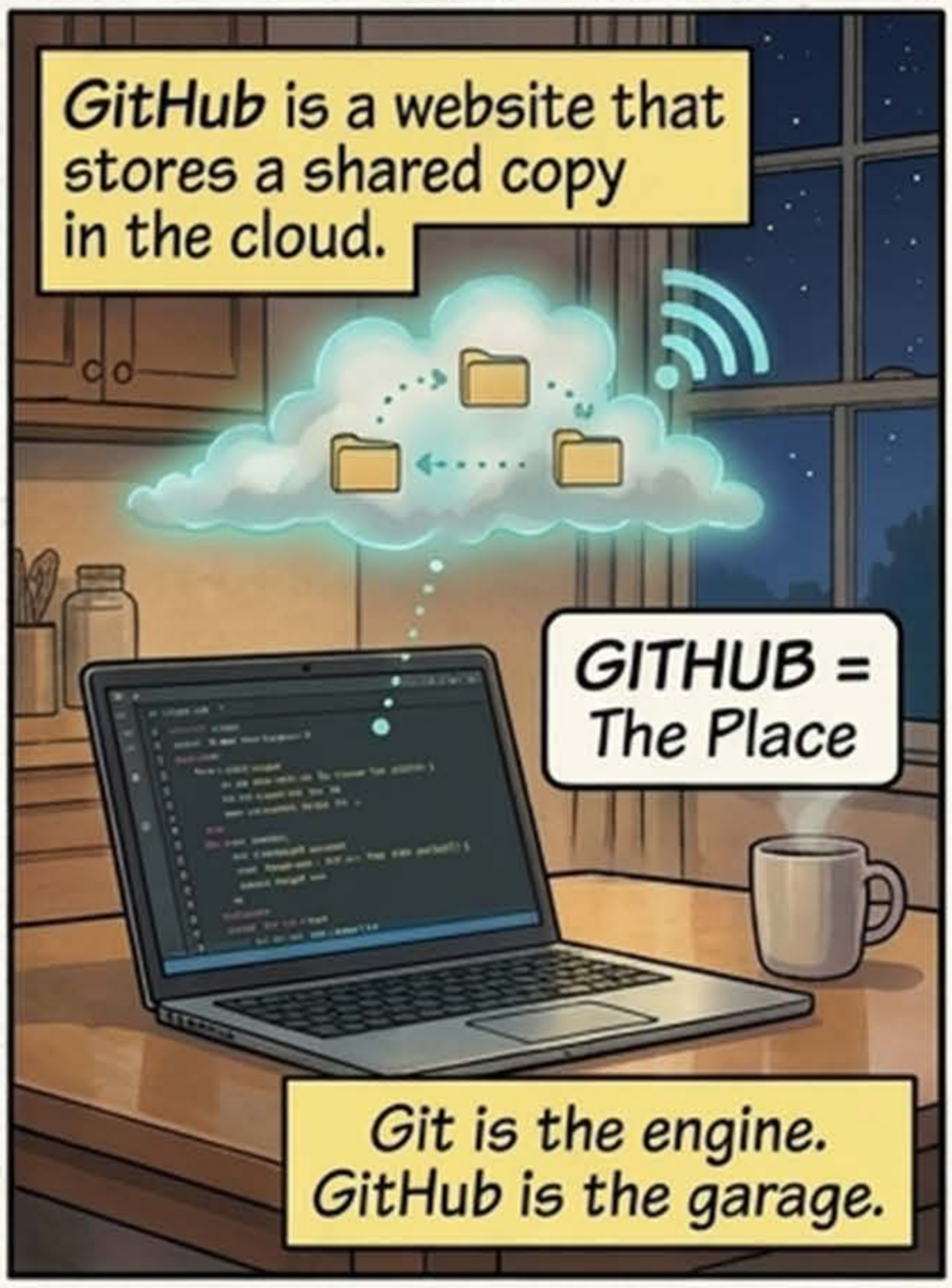
A learning opportunity. I have several others queued up.

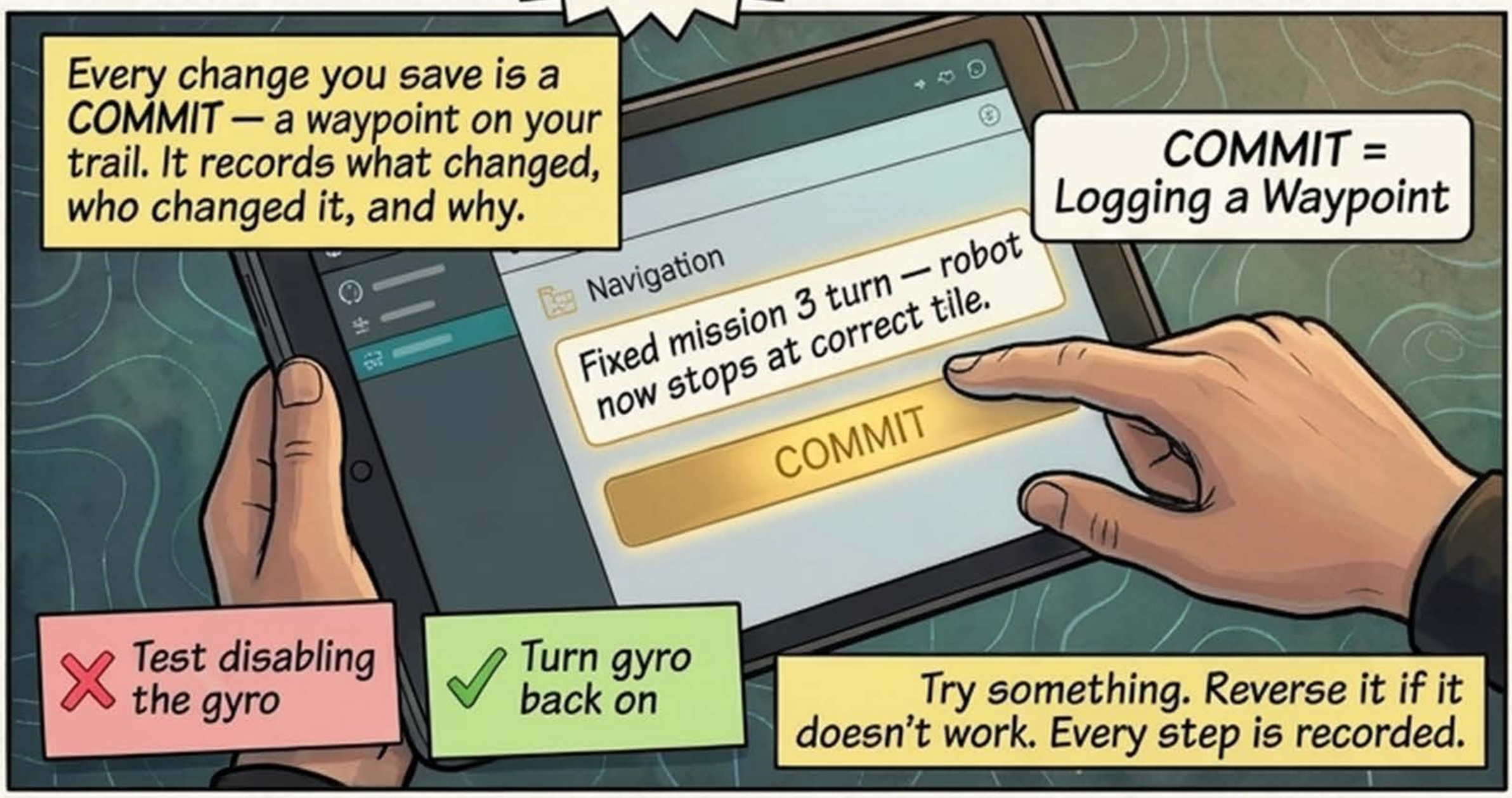
Every explorer knows: if you don't log your trail, you can lose your way.

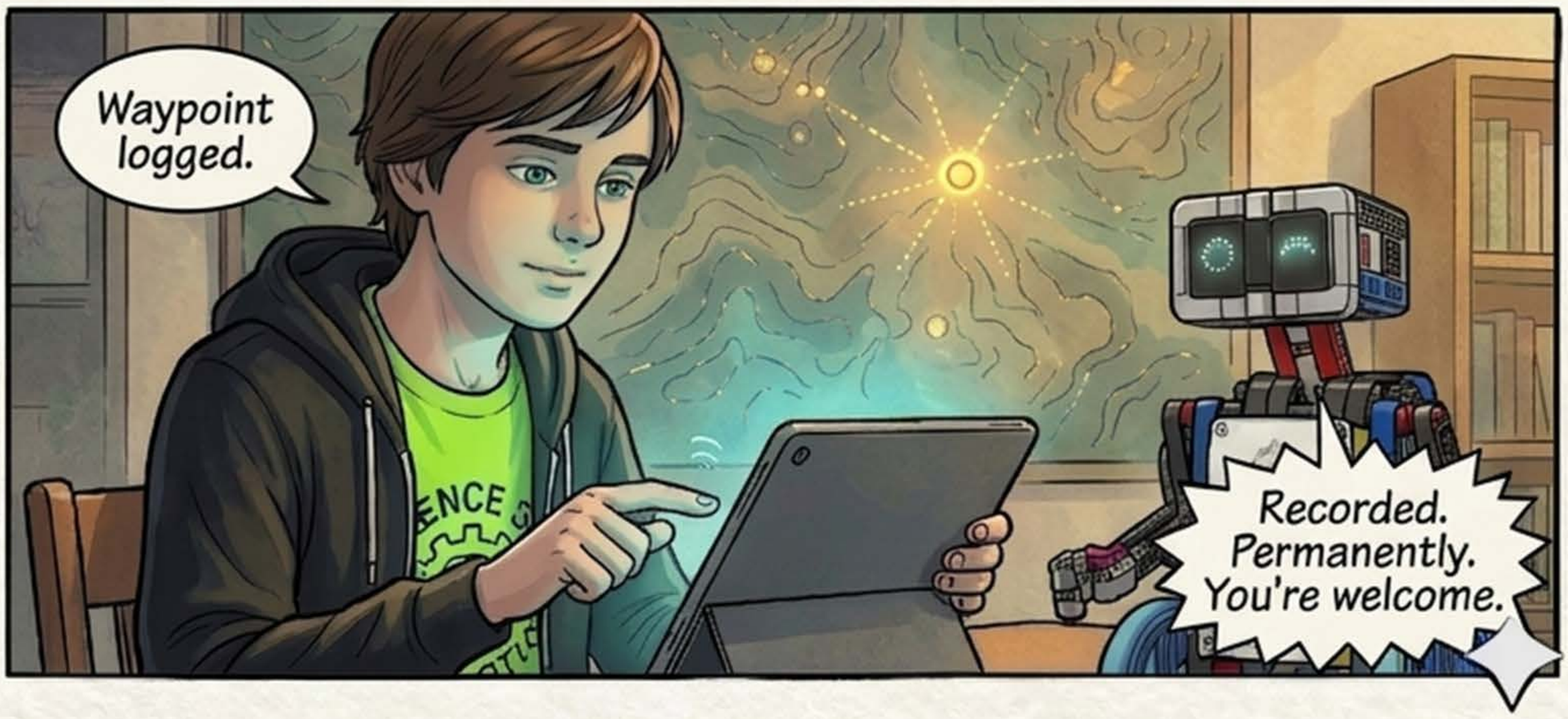
THE TOOL, THE PLACE, THE FIRST WAYPOINT

Git & GitHub / 1st Commit



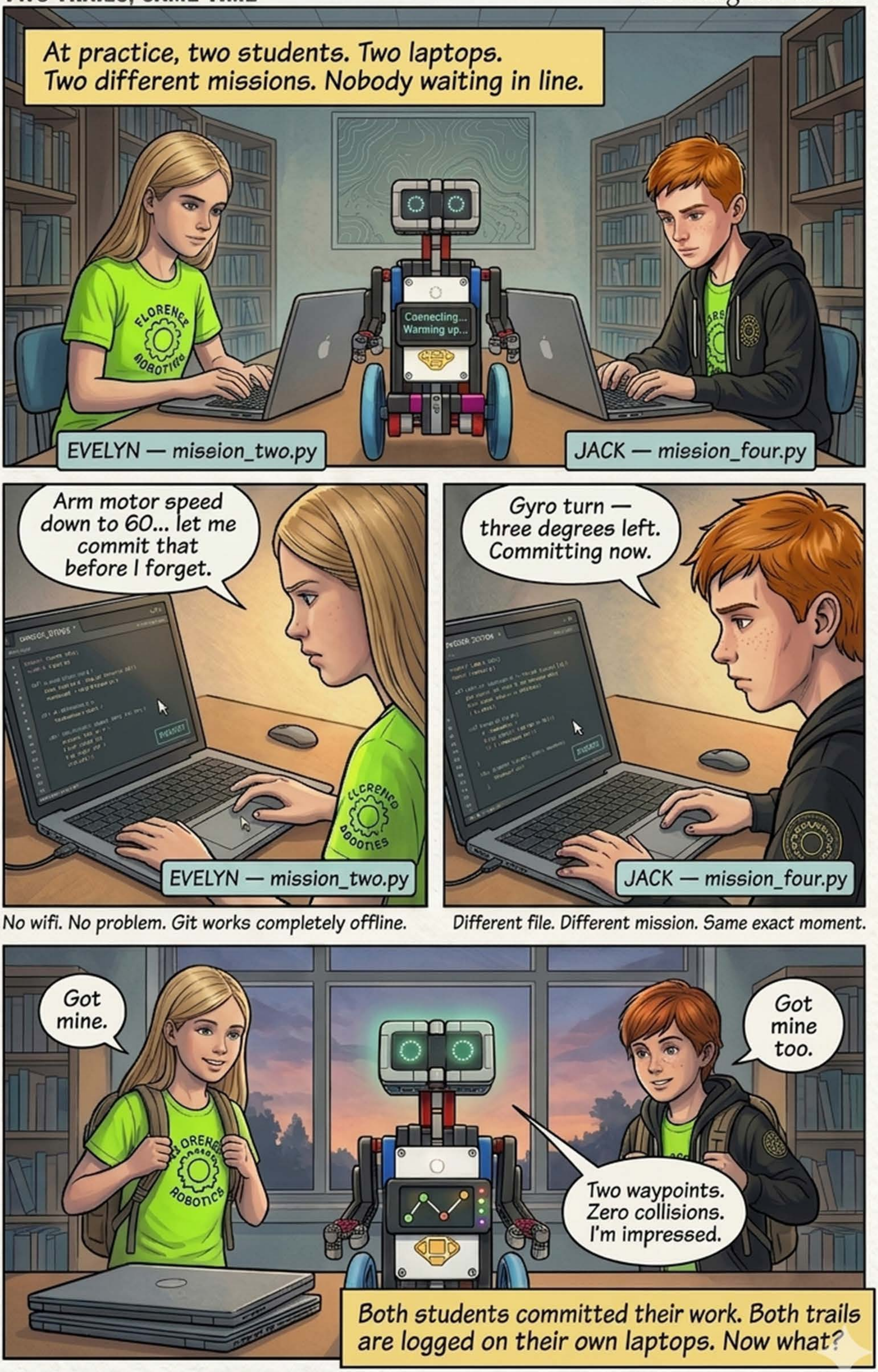






TWO TRAILS, SAME TIME

Working in Parallel

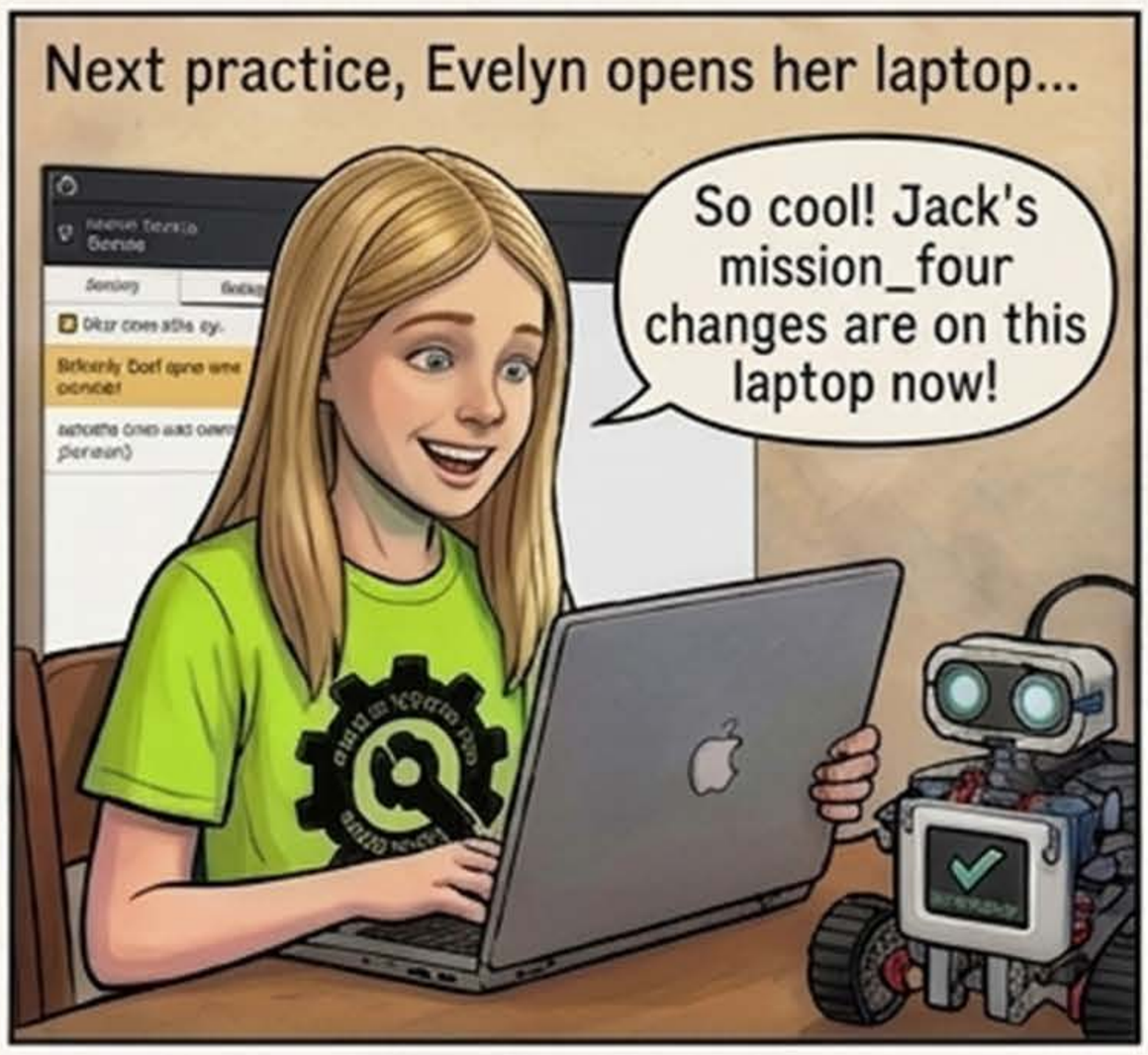


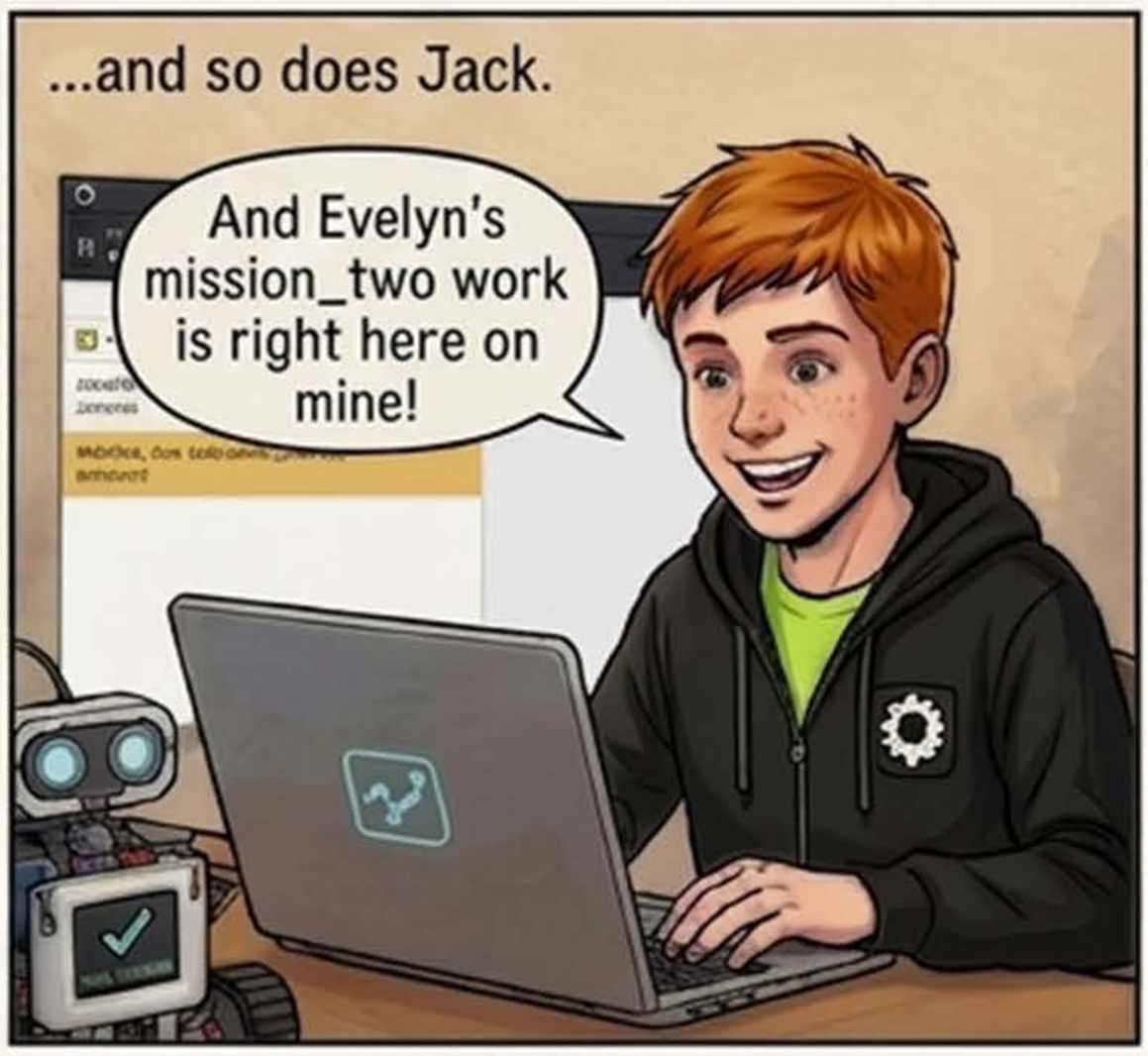
No wifi. No problem. Git works completely offline.

Different file. Different mission. Same exact moment.

TRAIL MAPS GET SYNCRONIZED

Code Push Integrates All Mission Changes



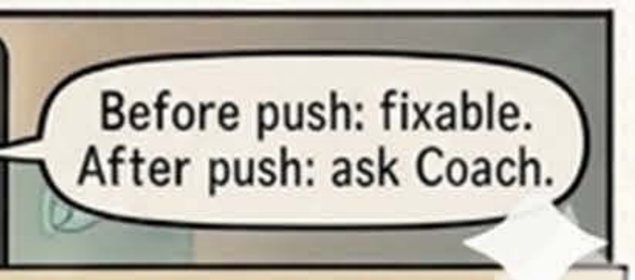


That's the magic. You commit. Coach pushes. Coach pulls. You arrive at the next session and your teammate's work is aiready there.

YOU COMMIT → COACH PUSHES → COACH PULLS → EVERYONE IN SYNC

PULL = Getting the Full Map Update

Made a mistake before Coach pushed? Right-click any commit, choose Undo. You retrace your steps. No harm done.



UNDO = Retracing Your Steps

THE TRAIL LOG NEVER LIES

The History tab shows every commit — who, when, and why. This is your trail log.

GitHub Desktop

History

d2a1b7 · Josie · Wednesday 4:42pm
mission 1: fixed base return path — was short by 15cm

c9f5a3 · Eleanor · Wednesday 4:18pm
mission 4: gyro turn left +3 degrees, now hits target

b8d4e2 · Jack · Wednesday 4:01pm
mission 3: arm motor speed 80→60, was overshooting

a3f2c1 · Coach · Wednesday 3:30pm
initial base code from chapter

Viewing History / Qualifier Day

Tagging The Code After a Qualifier

After every competition — before any new code — the coach creates a tag. A permanent bookmark on the exact code that ran.

TAG = Marking the Summit

Position lagged. Permanently.

So we can always come back to this exact code if we need to.

Five rules keep everything running.

TRAIL RULES

1. Pull before every session. Start with the latest map.
2. One group per mission, per practice. Same file on two laptope = trouble.
3. Write a useful commit message. Your future self will thank you.
4. Push is the coach's job. Students commit, coaches push. Safety net.
5. Tag after every competition. The run is preserved forever.

✓ Useful	✗ Not helpful
miesion 3: arm mctor speed 80→60 was overshooting	Fixed stuff

These aren't restrictions — they're how explorers keep each other safe.

Every waypoint we logged — it's all here.

Ready when you are, team.

The system is in place. The trail is clear. Now — go explore.

Workflow for Coding Missions

YOUR EXPLORER CHECKLIST.

- ☐ Write your code
- ☐ Test it on the robot
- ☐ Open GitHub Desktop
- ☐ Write a clear commit message
- ☐ Click COMMIT
- ☐ Tell your coach you're done
- ☐ Coaches handle the rest.

Your first commit starts right now. Open your mission file. Make a small change. Log your waypoint. You're an explorer.

Bolton Robotics · FSS FLL · github.com/fssfll/spike_basecode

D.A.L.E. – DATA ACQUISITION & LEARNING ENGINE

Why Explorers?

A note on the trail explorer metaphor to understand Git & GitHub.

You just read a comic about explorers, waypoints, and a robot named DALE. None of that is what GitHub actually looks like on a screen. So why tell the story this way?

Because explorers and software teams are solving the same problem.

When explorers map unfamiliar territory, they can't afford to lose track of where they've been. They log waypoints. They share maps. They mark summits. They keep records detailed enough that anyone joining later can retrace every step. Not because rules say so — because losing the trail is genuinely dangerous.

Software teams face the same problem, just with code instead of terrain. Every change is a step into territory the team hasn't explored yet. Without a system that records every step, the team gets lost. Files disappear. Good ideas get overwritten. Nobody remembers why something changed three weeks ago. The thumb drive walks home in someone's backpack and doesn't come back.

Git and **GitHub** solve this the same way explorers do.

A **commit** is like logging a **waypoint**. You're marking a specific point on the trail — what you changed, when, and why. Anyone can find their way back to that exact spot. Future you will thank present you for writing a message that actually says something.

A **push** is like a radio call to base camp. Your laptop holds today's trail. When the coach pushes to GitHub, that trail gets reported back to the shared map. Now the whole team knows where you've been.

A **pull** is like getting the full map update. Before your next session, the coach pulls everyone's combined work down to every laptop. You start with the complete map — every teammate's progress, already on your screen.

A **tag** is like marking the summit. After a competition, the coach plants a flag on the exact code that ran. Months later, if something breaks, the team can hike back to that summit and see what changed.

The metaphor works because it isn't just decoration. The vocabulary is different, but **the discipline is identical**.

No explorer gets lost twice.

FLORENCE SAWYER ROBOTICS